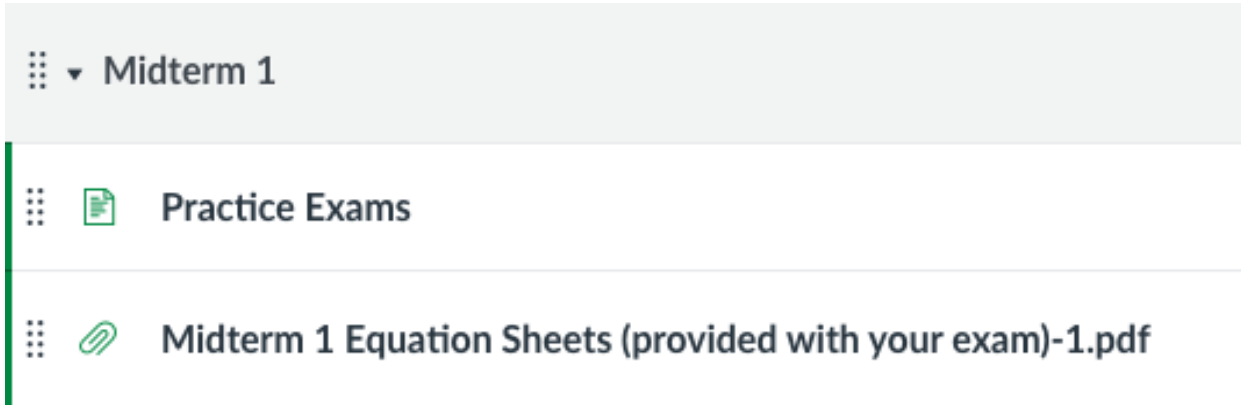


Lecture 12

- In-Class Practice Questions
- Some of Exam Questions

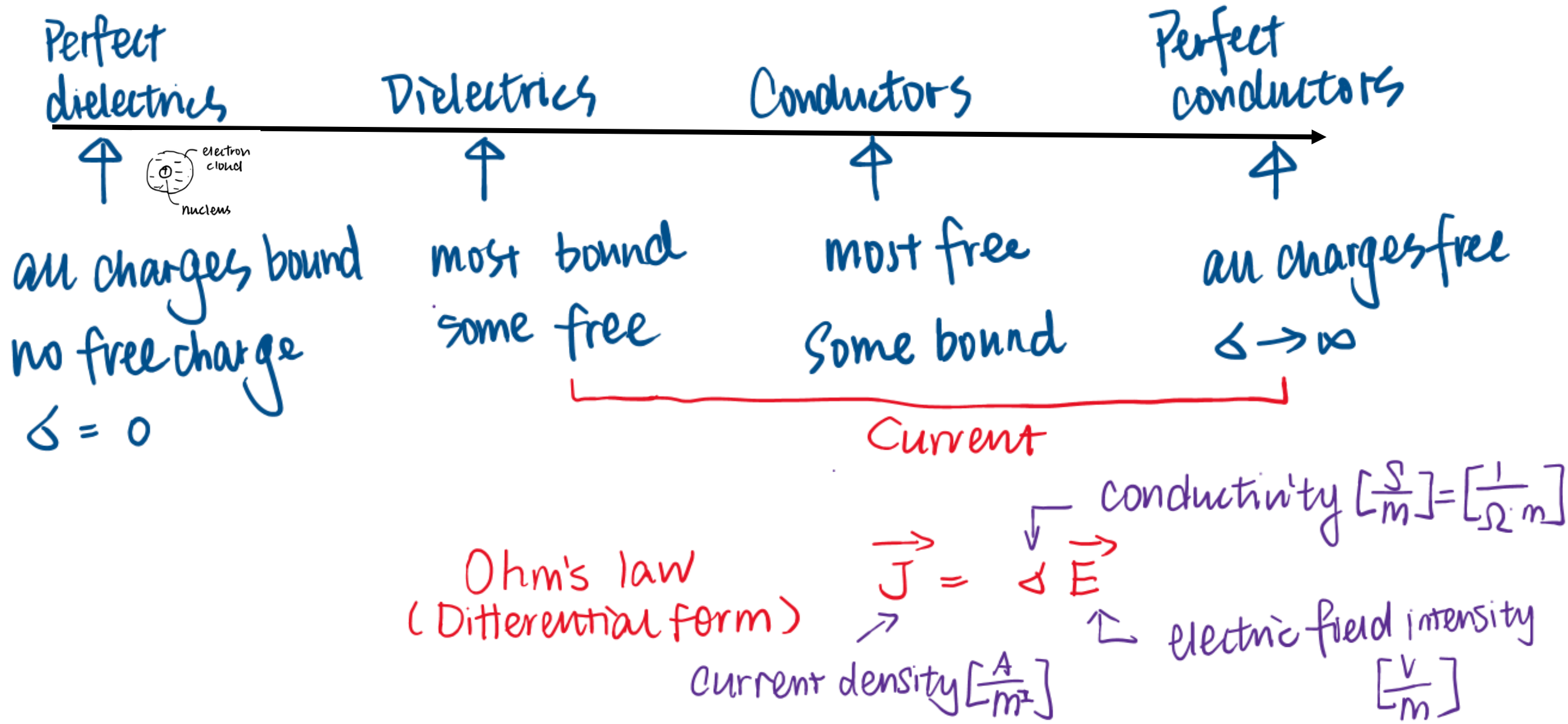
- 3 Midterm Exams + Solutions uploaded to Canvas, please practice.



- HW4 (only 5 relatively short questions) uploaded. Please practice before Midterm 1. Due date is Tuesday next week.
- Friday: Review session + Some exam examples

* Our Midterm 1: The focus is more on physical understanding, and the calculations should be quite straightforward. If you find yourself getting stuck on a calculation, you may want to try a different approach.

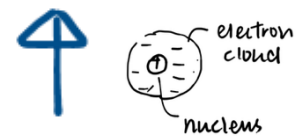
Depending on whether the material contains **free charges** that can readily move in response to an applied electric field, we classify them into two broad categories: **dielectrics** and **conductors**.



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Perfect dielectrics

Perfect conductors



all charges bound
no free charge
 $\epsilon = 0$

all charges free
 $\epsilon \rightarrow \infty$

No current

Have current, zero resistivity

Have electric field + polarization field

Inside PEC, electric field = 0

electric flux density [$\frac{C}{m^2}$]
(free charge)

polarization field [$\frac{C}{m^2}$]
(bound charge displacement)

electric field intensity [$\frac{V}{m}$]

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

In the presence of an external electric field, good dielectrics polarize

good conductors screen/shield

Review: Perfect conductor (PEC) in electrostatic equilibrium $(\sigma \rightarrow \infty)$

Note:-

$$\textcircled{1} \quad \vec{J} = \sigma \vec{E} \quad \vec{E} = \frac{\vec{J}}{\sigma} \rightarrow 0 \quad \text{Inside of PEC, } \vec{E} = 0$$

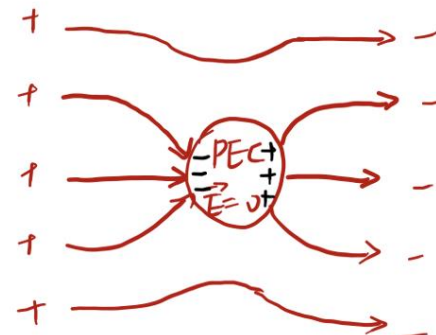
$$\vec{B} = 0 \quad \rightarrow \quad \nabla \cdot \vec{D} = \rho_v$$

② Inside of PEC, net volume charge density ρ_v is 0 everywhere.

- Where do the free charges go?
- The free charges distribute themselves to the surface, giving rise to the surface charge density ρ_s .

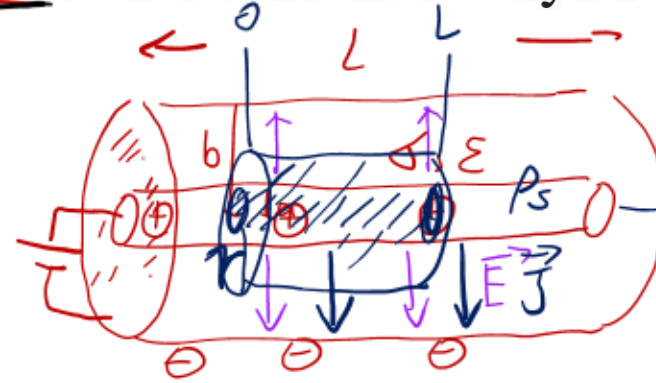
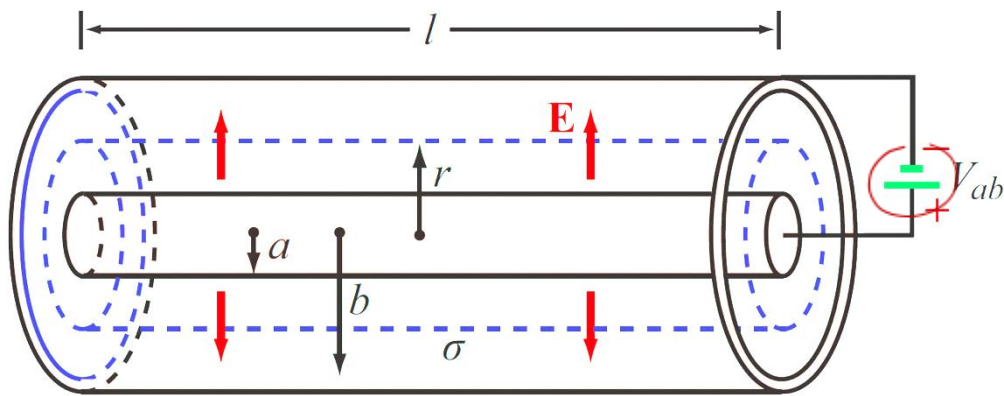
③ PEC is equipotential surface. $A \text{---} \vec{E}=0 \text{---} B \quad V_A = V_B$

④ $\vec{E}_{\text{external}}$ is always normal to PEC surface



Practice: Ohm's law

The radii of the inner and outer conductors of a coaxial cable of length l are a and b , respectively (**Fig. 4-17**). The insulation material has conductivity σ . Obtain an expression for G' , which is the conductance per unit length of the insulation layer.

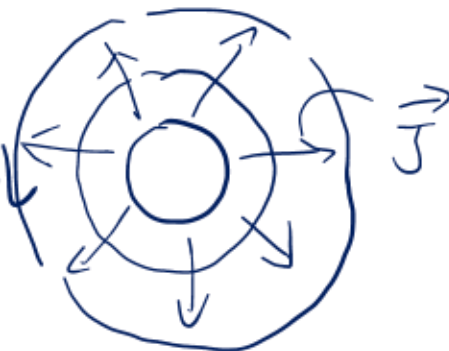


$$\vec{J} = \sigma \vec{E}$$

$$\vec{E} = \hat{r} E_r$$

$$D_r \cdot 2\pi r \cdot L = Q_{enc} = \rho_s \cdot 2\pi a \cdot L$$

$$D_r = \rho_s \frac{a}{r}$$



$$\vec{E} = \hat{r} \frac{D_r}{\epsilon} = \hat{r} \frac{\rho_s a}{r \epsilon}$$

$$G' = \frac{G}{L} = \frac{2\pi \sigma}{\ln(\frac{b}{a})}$$

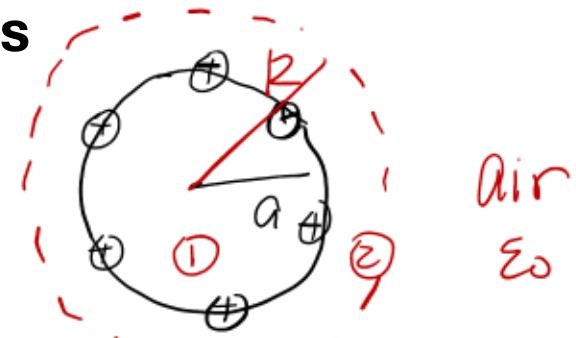
$$G = \frac{I}{V} = \frac{\frac{\sigma \rho_s a}{\epsilon} \cdot 2\pi L}{\frac{\rho_s a}{\epsilon} \ln(\frac{b}{a})} = \frac{\sigma 2\pi L}{\ln(\frac{b}{a})}$$

$$I = \int \vec{J} \cdot d\vec{s} = \int_{z=0}^L \int_{\phi=0}^{2\pi} \sigma \vec{E} \cdot \hat{r} r d\phi dz = \int_{z=0}^L \int_{\phi=0}^{2\pi} \sigma \left(\frac{\rho_s a}{r \epsilon} \right) r d\phi dz = \frac{\sigma \rho_s a}{\epsilon} \cdot 2\pi L$$

$$V = \int_+^- \vec{E} \cdot \hat{r} dr = \int_{r=a}^b \frac{\rho_s a}{r \epsilon} dr = \frac{\rho_s a}{\epsilon} \ln \frac{b}{a}$$

Exam questions related to electrical properties of materials

Charge Q has been placed on a metallic sphere of radius a (i.e., the sphere is made of an excellent conductor). The surrounding medium is air (relative permittivity $\epsilon_r = 1$).



Region ① $R < a$ $\vec{E} = 0$ $\vec{D} = 0$

Region ② $R > a$

$$\oint_S \vec{D} \cdot d\vec{S} = Q_{enc}$$

$$D_R \cdot 4\pi R^2 = Q$$

$$D_R = \frac{Q}{4\pi R^2}$$

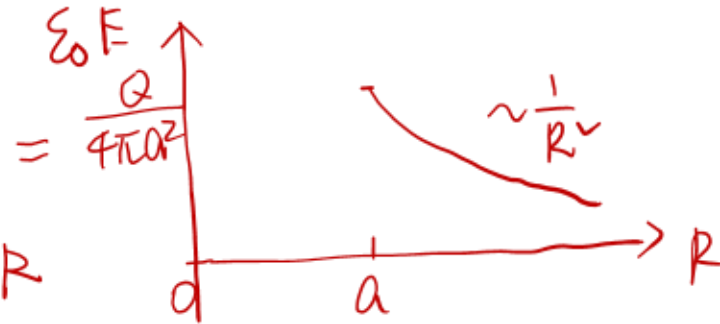
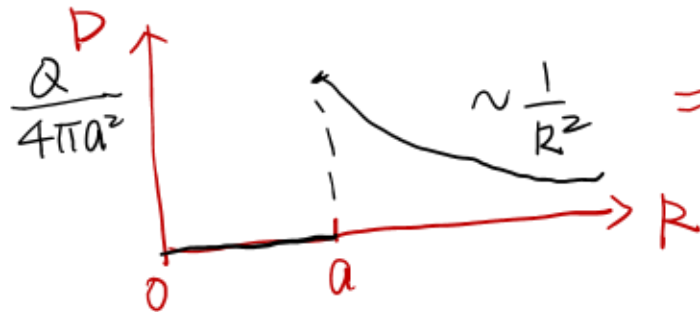
a) Use Gauss's law to calculate the electric flux density $\vec{D}$ and electric field intensity $\vec{E}$ everywhere in space. Graph $D(R)$ vs R and $\epsilon_0 E(R)$ vs R . (Hint: The sphere is metallic. That should guide your assumptions about how the charge is distributed.)

b) A thick dielectric spherical shell is added to the system (i.e., in addition to the charged sphere from before). The dielectric extends from $R = a$ to $R = b$, and has relative permittivity of $\epsilon_r = 2$. Calculate and graph $D(R)$ vs R , $\epsilon_0 E(R)$ vs R , and $P(R)$ vs R .

c) Calculate the potential difference $V(a) - V(b)$ for the setup described in part (b) (i.e., both the conductor and the dielectric shell together).

$$\vec{D} = \hat{R} \frac{Q}{4\pi R^2}$$

$$\vec{E} = \hat{R} \frac{Q}{4\pi \epsilon_0 R^2}$$

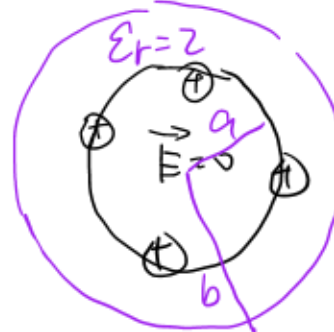


Exam questions related to electrical properties of materials

$$V(a) - V(b) = - \int_b^a \vec{E} \cdot \hat{R} dR$$

Charge Q has been placed on a metallic sphere of radius a (i.e., the sphere is made of an excellent conductor). The surrounding medium is air (relative permittivity $\epsilon_r = 1$).

a) Use Gauss's law to calculate the electric flux density $\vec{D}$ and electric field intensity $\vec{E}$ everywhere in space. Graph $D(R)$ vs R and $\epsilon_0 E(R)$ vs R . (Hint: The sphere is metallic. That should guide your assumptions about how the charge is distributed.)



$$= - \int_b^a \frac{Q}{8\pi\epsilon_0 R^2} dR$$

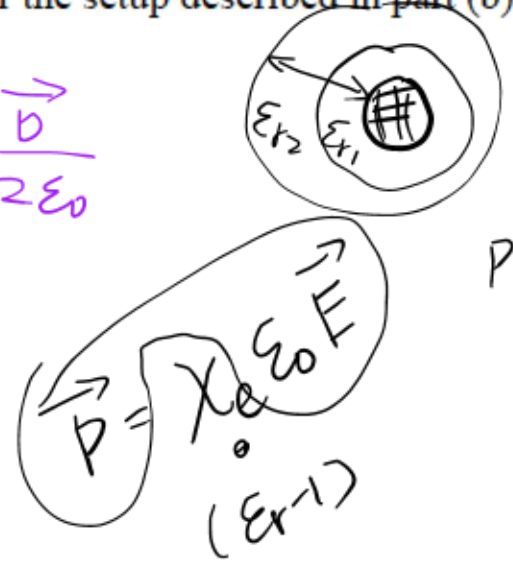
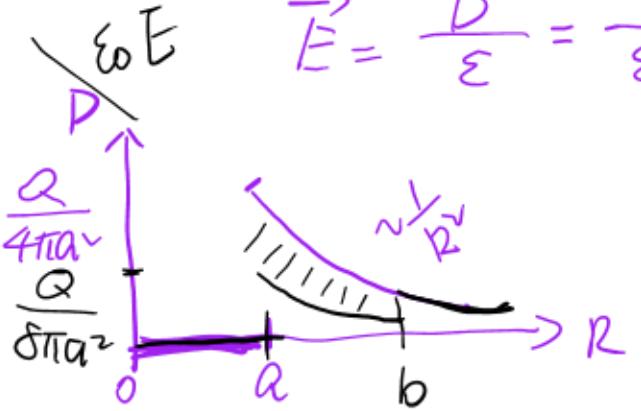
$$= - \frac{Q}{8\pi\epsilon_0} \left(-\frac{1}{R} \right) \Big|_b^a$$

b) A thick dielectric spherical shell is added to the system (i.e., in addition to the charged sphere from before). The dielectric extends from $R = a$ to $R = b$, and has relative permittivity of $\epsilon_r = 2$. Calculate and graph $D(R)$ vs R , $\epsilon_0 E(R)$ vs R , and $P(R)$ vs R .

$$\vec{D} = \begin{cases} 0 & R < a \\ \hat{R} \frac{Q}{4\pi R^2} & R > a \end{cases} = \frac{Q}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

c) Calculate the potential difference $V(a) - V(b)$ for the setup described in part (b) (i.e., both the conductor and the dielectric shell together).

$$\vec{E} = \frac{\vec{D}}{\epsilon} = \frac{\vec{D}}{\epsilon_0 \epsilon_r} = \frac{\vec{D}}{2\epsilon_0}$$



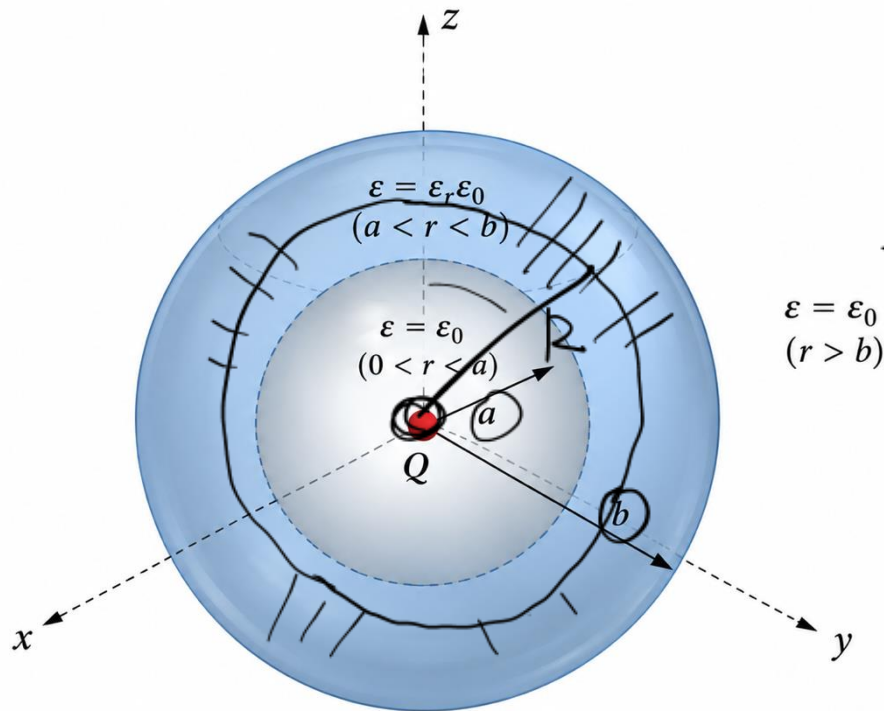
$$\Rightarrow \vec{E} = \begin{cases} 0, & R < a \\ \hat{R} \frac{Q}{8\pi\epsilon_0 R^2}, & a < R < b \\ \hat{R} \frac{Q}{4\pi\epsilon_0 R^2}, & R > b \end{cases}$$



Practice: Dielectrics, polarization field

Consider the picture below as a spherical dielectric shell, where $\epsilon = \epsilon_r \epsilon_0$ for $a < r < b$ and $\epsilon = \epsilon_0$ for $0 < r < a$. A charge Q is placed at the center of the shell.

Find the polarization field $\vec{P}$ for $a < r < b$.



- ~~Consider just part (a)~~, which asks you for the polarization field $\vec{P}$. What equation will we use to find $\vec{P}$? You should see that we can write $\vec{P}$ as a function of only ϵ_r and $\vec{D}$.
- Determine $\vec{D}$. Try doing this in just two lines!
- Determine $\vec{P}$.
- Sketch plots of $D(R)$, $E(R)$, and $P(R)$ as a function of R .

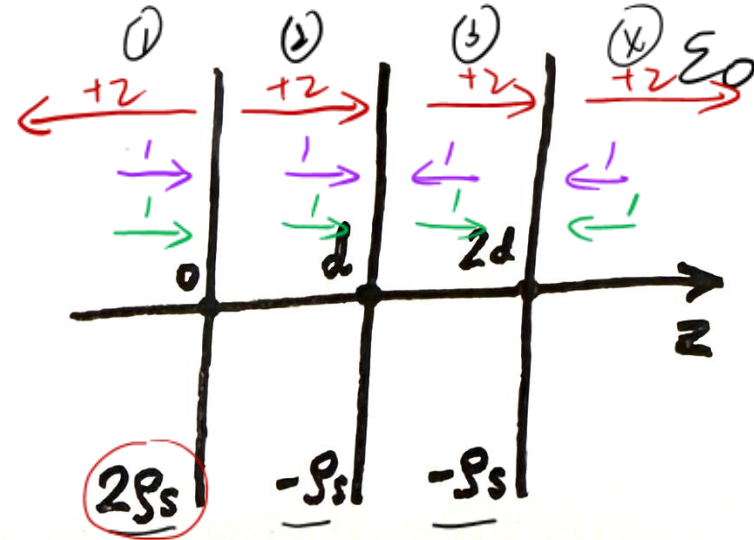
$$a) \quad \vec{P} = (\epsilon_r - 1) \epsilon_0 \frac{\vec{D}}{\epsilon_r} = \frac{\epsilon_r - 1}{\epsilon_r} \vec{D} \quad \leftarrow$$

$$b) \quad D_R \cdot 4\pi R^2 = Q \quad D_R = \frac{Q}{4\pi R^2} \quad \vec{D} = \hat{R} \frac{Q}{4\pi R^2}$$

$$c) \quad \vec{P} = \hat{R} \frac{\epsilon_r - 1}{\epsilon_r} \frac{Q}{4\pi R^2}$$

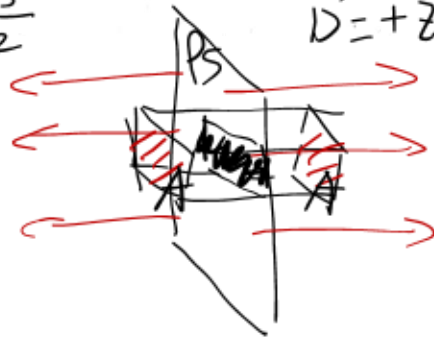
Exam questions related to electrical properties of materials

Problem 3: Three infinite parallel planes have surface charge densities $2\rho_s$, $-\rho_s$, $-\rho_s$ ($\rho_s > 0$). They are normal to the z -axis and placed at $z=0$, $z=d$, and $z=2d$, respectively, as in the figure. (Read carefully. This is NOT the same as the practice midterm!). Assume the planes are in vacuum (permittivity ϵ_0).



- a) Find $\vec{E}$ everywhere in space. Graph $E(z)$ versus z .
- b) Calculate and graph the electric potential $V(z)$. Assume $V(0) = 0$.

$$\vec{D} = -\hat{z} \frac{\rho_s}{2}$$



$$\vec{D} = +\hat{z} \frac{\rho_s}{2}$$

$$\oint_S \vec{D} \cdot d\vec{S}$$

$$2D_z \cdot A = \rho_s A$$

$$D_z = \frac{\rho_s}{2}$$

$$\rho_s \Rightarrow |\vec{D}| = \frac{\rho_s}{2}$$

Region ① $z < 0$, $\vec{E} = 0$

② $0 < z < d$, $\vec{E} = 4 \frac{\rho_s}{2\epsilon_0} \hat{z}$

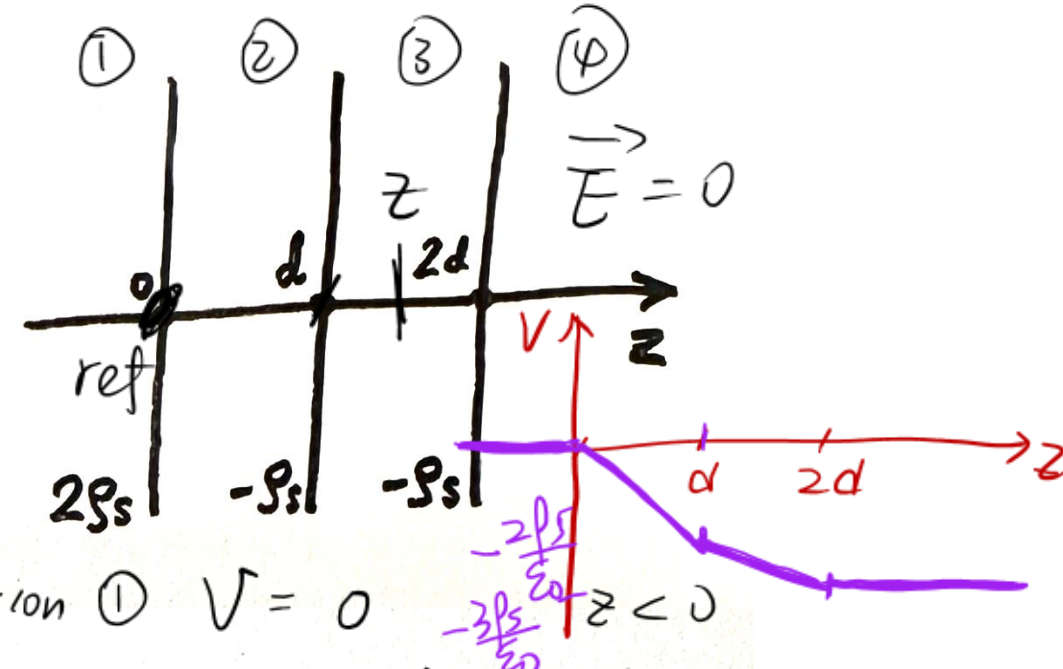
$$= \hat{z} \frac{2\rho_s}{\epsilon_0}$$

③ $d < z < 2d$, $\vec{E} = \hat{z} 2 \frac{\rho_s}{2\epsilon_0} = \hat{z} \frac{\rho_s}{\epsilon_0}$

④ $z > 2d$, $\vec{E} = 0$

Exam questions related to electrical properties of materials

Problem 3: Three infinite parallel planes have surface charge densities $2\rho_s$, $-\rho_s$, $-\rho_s$ ($\rho_s > 0$). They are normal to the z -axis and placed at $z=0$, $z=d$, and $z=2d$, respectively, as in the figure. (Read carefully. This is NOT the same as the practice midterm!). Assume the planes are in vacuum (permittivity ϵ_0).



a) Find $\vec{E}$ everywhere in space. Graph $E(z)$ versus z .

b) Calculate and graph the electric potential $V(z)$. Assume $V(0)=0$.

$$\vec{E} = \begin{cases} 0 & z < 0 \\ \hat{z} \frac{2\rho_s}{\epsilon_0} & 0 < z < d \\ \hat{z} \frac{\rho_s}{\epsilon_0} & d < z < 2d \\ 0 & z > 2d \end{cases}$$

Region ④ $V = -\frac{3\rho_s}{\epsilon_0}d$

$$\begin{aligned} \text{Region ②} \quad V &= -\int_{z=0}^z \frac{2\rho_s}{\epsilon_0} dz \\ &= -\frac{2\rho_s}{\epsilon_0} z \\ \text{Region ③} \quad V &= -\int_{z=0}^d \vec{E}_2 \cdot d\vec{l} - \int_{z=d}^z \vec{E}_3 \cdot d\vec{l} \\ &= -\frac{2\rho_s}{\epsilon_0}d - \left(\frac{\rho_s}{\epsilon_0}z - \frac{\rho_s}{\epsilon_0}d \right) \\ &= -\frac{\rho_s}{\epsilon_0}d - \frac{\rho_s}{\epsilon_0}z \end{aligned}$$